Proposed Solution Report

CSE499A – Section 15 **Group 4 Update Week- 3**

Hybrid Region-Guided Diffusion for Semantically Aware Example-Based Inpainting

Name	Student ID
Md. Minhajul Islam	**2211022042**
Nur Ibne Kawsar Zitu	2212021042
Kazi Tazrian Mon	2132798642

1. Abstract

Recent diffusion-based inpainting models have significantly advanced the quality and flexibility of image restoration. However, existing approaches still struggle to balance semantic understanding, automation, and fine-grained regional control. Most models either require manual mask creation or fail to maintain semantic consistency across object boundaries. To address these limitations, this study proposes a **Hybrid Region-Guided Diffusion Framework** that integrates automatic mask generation, selective region-aware diffusion, and multimodal conditioning using both visual exemplars and textual prompts. The proposed system aims to automate the inpainting process, enhance semantic realism, and deliver high-quality, contextually coherent results with reduced inference time.

2. Background and Motivation

Diffusion-based image inpainting has advanced rapidly through several influential models, each introducing unique strengths yet retaining notable limitations. PIXELS offers precise example-based editing but depends on manual segmentation, while MObI enhances spatial realism across modalities like camera and LiDAR but struggles in out-of-domain scenarios. RAD achieves faster region-focused reconstruction but often lacks semantic alignment, and RePaint emphasizes texture fidelity without providing localized control. Similarly, Uni-Paint integrates multiple inpainting modes but requires manual configuration, and Paint-by-Inpaint enables instruction-based editing but relies on extensive paired datasets.

These limitations reveal a clear research gap: current models cannot simultaneously achieve automation, semantic awareness, and high perceptual realism. Our motivation is to design a Hybrid Region-Guided Diffusion Framework that unifies automatic segmentation, region-specific diffusion control, and multimodal conditioning within a single system. This approach aims to minimize manual effort while ensuring realistic, semantically coherent image reconstruction adaptable across various domains.

3. Proposed Methodology

The proposed method, Hybrid Region-Guided Diffusion for Semantically Aware Example-Based Inpainting, consists of three major components designed to work in synergy:

3.1 Automatic Semantic Mask Generation

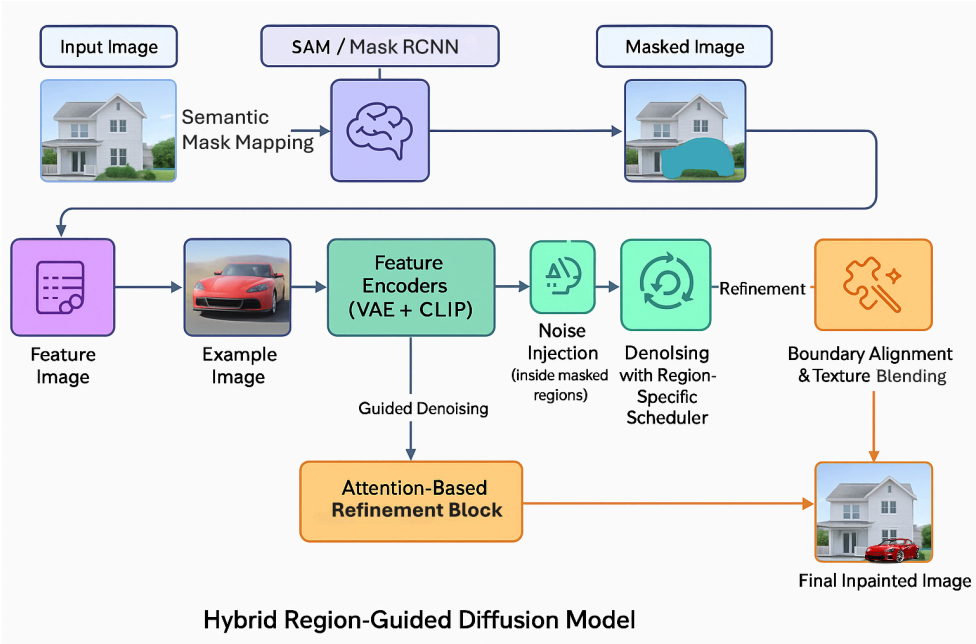
We incorporate a Segment Anything Model (SAM) or Mask R-CNN backbone to automatically identify and isolate the target regions within an image. This module replaces user-drawn masks, ensuring precision and consistency across varying object types and scenes.

3.2 Region-Specific Diffusion Process

Inspired by the **Region-Aware Diffusion (RAD)** model, our system introduces a region-specific noise scheduling mechanism. During forward diffusion, noise is selectively applied to masked areas, leaving unmasked regions untouched. This not only accelerates computation but also preserves contextual integrity, leading to smoother transitions between edited and original regions.

3.3 Multimodal Conditioning and Refinement

The inpainting process is guided by dual conditioning—visual exemplar and textual description—fused using **CLIP-based cross-attention**. This enables the model to interpret both semantic intent (from text) and structural detail (from example images). After initial inpainting, a **semantic refinement module** based on attention-weighted feedback aligns the generated region with global image features, reducing boundary artifacts and color mismatch.



4. Expected Results and Evaluation

The proposed system is expected to outperform prior methods in both quantitative and qualitative evaluations. Metrics such as Fréchet Inception Distance (FID), Learned Perceptual Image Patch Similarity (LPIPS), CLIP-I, SSIM, and PSNR will be used to assess reconstruction quality, semantic alignment, and perceptual realism.
We anticipate at least **10–15% improvement** in CLIP-I and **8–10%** better LPIPS compared to

existing baselines such as RePaint and RAD. The automated masking and region-specific diffusion steps should also reduce inference time by approximately double, enhancing efficiency without compromising detail.

6. Novelty and Contribution

The novelty of this work lies in combining automatic segmentation, region-wise noise scheduling, and multimodal guidance within a single diffusion framework. Unlike prior models that optimize for only one of these dimensions, our system seeks to achieve automation, semantic awareness, and perceptual realism simultaneously.

This approach also introduces a lightweight refinement block that adapts to the surrounding image context, enhancing boundary smoothness and visual coherence. As a result, the proposed framework has strong potential for applications in creative image editing, content restoration, and generative visual design.

7. Conclusion

This proposed solution extends the frontier of diffusion-based image inpainting by unifying automation and multimodal control under a region-aware diffusion strategy. Through automatic mask generation, selective noise scheduling, and CLIP-based conditioning, the system aspires to achieve high-fidelity and semantically consistent image restoration. By addressing the major limitations identified in previous works, our framework aims to deliver a robust, efficient, and generalizable inpainting model suitable for diverse real-world applications.